Pranav Hemanth

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Education

PES University — **Bangalore, India**
Bachelor's in Technology, Computer Science — *Aug 2023 – June 2027*
CGPA: 9.24/10
Relevant Courses: Machine Learning, Databases, Software Engineering, Algorithms and Data Structures, Operating Systems, Network and Information Security, Cryptography, Digital Forensics, Web Technologies, Computer Networks, Computer Architecture and Organisation, Data Science, Calculus and Discrete Mathematics
Academic Highlights: Top 10% of class, 5× Hackathon winner, Research in Cybersecurity, Systems and Machine Learning

Publications & Submissions

[3]: **P. Hemanth**, K. Pawar, "Four Evaluation Confounds in Multi-Stage Knowledge-Graph Intrusion Detection," *under review*, ANUBIS @ ESORICS 2026.

[2]: S. Saha, **P. Hemanth**, "Skilldex: A Package Manager for Portable, Consistent Agent Skills in the Open SKILL.md Standard," *under review*, EMNLP 2026 System Demonstrations. arXiv:2604.16911.

[1]: **P. Hemanth**, S. Saha, "Conversation Tree Architecture: A Structured Framework for Context-Aware Multi-Branch LLM Conversations," arXiv preprint arXiv:2603.21278, 2026.

Professional Experience

Centre for Cybersecurity, Trust and Reliability, IIT Madras — **Chennai, India**
Research Intern — *Jun 2026 – Present*
- Developing an internal tool for Blockchain Forensics investigations.

Pandemonium Research — **Bangalore, India**
Co-founder and Researcher — *Mar 2026 – Present*
- Co-founded an independent engineering and research organization spanning advanced AI systems, distributed systems, cybersecurity, developer tooling, and fintech infrastructure.
- Shipped **Skilldex**[2] as an open-source CLI and community-driven registry with package manager distribution (Homebrew, Scoop), enabling standardized agent skill discovery across engineering teams.

Cyber Manthan Center, IIIT Hyderabad — **Hyderabad, India**
Cybersecurity and Machine Learning Research Intern — *Dec 2025 – Mar 2026*
- Built a low-resource, agentic framework for SOC alert triage across Suricata, Zeek, Wazuh, and OSSEC, delivering MISP-grounded anomaly scores and reasoned mitigation recommendations to analysts.

C3iHub, IIT Kanpur — **Kanpur, India**
Cybersecurity Research and Development Intern — *Jun 2025 – Sept 2025*
- Developed a unified heterogeneous attack graph system[3] with knowledge graph embeddings for multistage threat detection.
- Engineered and operationalized a custom alert correlation framework, achieving **500× to 10,000×** alert reduction across Suricata, Zeek, Wazuh, and OSSEC.

buildspace Nights & Weekends S5 — **Remote**
Project Intern — *Jun 2024 – Jul 2024*
- Proposed a novel imperceptible and tamper resistant digital image watermarking technique to combat visual misinformation.
- Applied steganography, cryptography, and machine learning concepts to design a watermarking pipeline.
- Led project planning and implementation, delivering documented code and research outcomes at buildspace Demo Day.

Helpcreate
Chief Technology Officer, Co-Founder — *Oct 2020 – Aug 2021*
- Co-founded a platform enabling innovators to transform ideas into businesses via a community-centric web app.
- **Achievement: 3rd Place**, Eureka! Junior @ **IIT Bombay**. Secured seed funding.

Academic Roles & Achievements

PES University — **Bangalore, India**
Teaching Assistant, Operating Systems — *Jan 2026 – May 2026*
- Curating study material and problem sets; evaluating students for projects on systems.

Teaching Assistant, Automata Formal Languages and Logic (AFLL) *Aug 2025 – Dec 2025*

- Created study material and problem sets; guided students for lexer projects and formal language modeling.

Subject Matter Expert, Applied Cryptography (PESU I/O) *Sept 2025 – Dec 2025*

- Developed and taught classical/modern cryptography modules; hosted CTF challenges integrating crypto, OSINT, forensics.

University of Oxford **Remote**

Machine Learning Summer School (OxML 2025) *Aug 2025*

Among the youngest participants. Attended intensive sessions on AI safety, generative AI, robotics, and representation learning.

BMRCL(Bangalore Metro Rail Corporation Limited) **Bangalore, India**

Security Assesment *Jul 2025*

Supported in verification of full scale VAPT assessment on all subdomains of and infrastructure of BMRCL leading to lawful disclosure of vulnerabilities.

Projects

SCIONMiniNet: Learned SCION architecture hands-on. Run a complete multi-AS network locally with Docker, explore multi-path routing, and experiment with path-aware networking inspired by ETH Zurich research.

ARMSim-Docker: Built a Dockerized ARM simulator environment to allow macOS users to run ARMSim for Microprocessor Control and Computer Architecture coursework without native installation. Adopted by peers for cross-platform development.

GPT-From-Scratch: Explored transformer internals by building a GPT model using PyTorch and nanoGPT, implementing attention mechanisms and training pipelines from fundamentals.

PolymorphicMalware: Educational security research project demonstrating polymorphic malware behavior including encryption, self-modification, and controlled infection for analysis.

TermBrowse: Terminal-based web browser in Python supporting search, text rendering, link navigation, bookmarking, and readability enhancements within the terminal.

AlgoTrader: Algorithmic trading bot using Alpaca's API with custom strategy logic to explore high frequency and machine learning based trading systems.

Hackathons, CTFs & Competitive Engineering

Indian Institute of Technology, Madras **Chennai, India**

Shaastra, IIT Madras CTFs and Challenges *Jan 2024 – Jan 2026*

Three time finalist across Reverse Engineering, Competitive Programming Potpourri, and Shaastra CTF.

Manipal institute of Technology (National)

MIST National OSINT Challenge *Apr 2025 and Nov 2025*

Ranked **32nd nationally** (Apr 2025) and **17th nationally** (Nov 2025) in national scale OSINT competitions.

Center for Cybercrime Investigation Training and Research (CCITR) - CID, Karnataka **Bangalore, India**

***4th Place**, CIDECODE 2.0* *Mar 2025*

IoTrace, a WearOS forensic data extraction and AI analysis system for law enforcement use.

Punjab National Bank | Indian Institute of Technology, Kanpur **Bangalore, India**

***Finalist**, PNB Cybersecurity Hackathon* *Dec 2024 - Mar 2025*

Built an end to end intrusion detection system to protect banking infrastructure against ransomware families such as Akira and LockBit.

PES University **Bangalore, India**

***1st Place** ×3 - Kodikon 3.0, Arithmania 3.0, Robowars* *Nov 2023 – Oct 2024*

R3ND3R, an automated React component rendering and test case generation framework (Kodikon 3.0, Embrione, Oct 2024); *Blockheads*, a blockchain based voting system for transparent and accessible elections (Arithmania 3.0, Shunya, Apr 2024); autonomous Python controlled line following robot built by Team 405 Found (Robowars, IEEE RAS, Nov 2023).

Skills

Programming: Python, C/C++, Go, JavaScript (Node.js, React.js, Next.js), HTML/CSS, PHP, SQL, Verilog, Assembly

ML/AI: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV

DevOps & Tools: Docker, Kubernetes, CI/CD, GitHub, AWS, GCP, DigitalOcean, Bash/zsh, Dockerfile

Security & Infra: RabbitMQ, Kafka, ELK, Wazuh, OSSEC, Suricata, Zeek, Dgraph, GraphQL, KQL, DQL, NetworkX